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RESEARCH ARTICLE

Neurophysiology of Exposure to Extreme Environments–Pressure, Temperature, & Microgravity

Effects of 48-h exposure to natural hypobaric hypoxia on surface-electromyography-based cocontraction indices

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Abstract

Hypobaric hypoxia (HH) affects the nervous system's ability to stabilize motor tasks, primarily through changes in neuromuscular activation. Previous studies have reported inconsistent findings regarding electromyographic responses under HH conditions, possibly due to a focus on individual muscle behaviors rather than intermuscular coordination. This study aimed to determine the effects of HH on surface electromyography-based cocontraction indices (CCIs) during a repetitive upper extremity task and to evaluate the impact of acute (<3 h) and prolonged (48 h) exposure. A cross-sectional study was conducted at 3,600 m above sea level, involving 12 healthy adults (5 males, 7 females). Surface electromyographic activity of the biceps and triceps brachii muscles was recorded during a repetitive reaching-retrieving task. Two CCIs were calculated. A significant effect of HH exposure time was observed, with higher CCIs after 48 h compared with acute exposure ($F_{1,44} = 4.172$; $P = 0.047$, $\eta_p^2 = 0.515$). No significant interactions between task phases or movement durations were found. Exposure to HH for 48 h significantly increases CCIs, suggesting compensatory neuromotor responses to HH. These findings highlight the utility of CCIs as markers of neuromuscular alterations during HH and provide insights into the strategies used by the nervous system under extreme conditions. Future studies should explore these responses over longer periods and across diverse motor tasks.

NEW & NOTEWORTHY A 48-h exposure to natural hypobaric hypoxia increases cocontraction indices, suggesting compensatory neuromotor adaptations. These findings highlight hypoxia-induced motor control changes and support the use of cocontraction indices as markers of neuromuscular adaptation in extreme environments such as high-altitude hypobaric hypoxia.

cocontraction index; electromyography; hypobaric hypoxia; hypoxia; motor control

INTRODUCTION

Hypobaric hypoxia (HH) is an environmental condition characterized by reduced atmospheric pressure, leading to a lower fraction of inspired oxygen (F_{IO_2}). Continuous acute HH exposure (i.e., without periods of normoxia) has been shown to impair human performance, affecting several physiological systems, including the nervous system (1–3). Compromised psychomotor skills observed under HH demand acute and chronic neuromuscular adaptations (4) to optimize processes related to joint stabilization during the execution of a motor task (5). One key neural mechanism involved in this adaptation is muscle activation, which can be assessed using surface electromyography (EMG).

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From a metabolic standpoint, acute HH exposure is associated with increased venous blood lactate and intramuscular H^+ concentrations (6), reflecting the inherent metabolic stress during HH, which could, in turn, affect muscle activation and coordination.

Interestingly, it has been found that the magnitude of EMG (7–9) does not differ (10, 11) and even decreases (12) under hypoxic conditions. These conflicting findings could be partially explained by the fact that EMG characteristics under HH conditions respond to individual muscle behaviors rather than the intermuscular coordination that regulates motor actions. Accordingly, from a neuromotor perspective, cocontraction indices (CCIs) allow for the analysis of the simultaneous coordinated activation of muscles on opposite sides of a joint (13). This approach enables the examination of the nervous system's neurophysiological strategies to regulate joint stability and enhance movement precision (14).

CCIs have proven to be an accurate tool for quantifying the dynamic interaction between agonist and antagonist muscle groups during motor tasks (15–17). Higher CCI values indicate a greater neuromuscular demand for the nervous system to stabilize the joint, which has been interpreted as an increased metabolic cost associated with motor control deficits (14, 18). Therefore, metabolic cost refers to an inferred neuromuscular demand associated with HH, rather than a directly measured physiological variable.

Despite their potential, there is no consensus on a standard method for calculating CCIs, nor on their reliability as markers of neuromotor performance under HH conditions. As a result, their utility as indicators of neuromotor performance in HH remains unclear. Furthermore, CCIs could play a valuable role in assessing human performance in extreme work environments, such as high-altitude mining operations. Workers exposed to HH often face neuromuscular challenges that may increase their risk of injury or reduce task efficiency. Monitoring CCIs may help detect motor control deficits and guide the implementation of targeted interventions to enhance joint stability and reduce metabolic costs. These applications align with the increasing focus on occupational safety and performance optimization in extreme conditions. Therefore, considering the limited evidence and controversial results, this study aims to determine the effects of HH on surface EMG-based cocontraction indices during a repetitive upper extremity task.

METHODS

Study Design and Ethics

This cross-sectional study was conducted in Socaire, located in the Antofagasta region of Chile, at an altitude of over 3,600 m above sea level (19). This research was conducted in accordance with the Helsinki Declaration on biomedical research involving human subjects. The study protocol was reviewed and approved by the Scientific Research Ethics Committee of the University of Antofagasta (CEIC-UA, File No. 457/2023).

Participants

Healthy adult volunteers were recruited for the study. All participants were lowlanders with no previous exposure to high-altitude working shifts. Exclusion criteria included presence of low back pain or upper extremity injuries; musculoskeletal or cardiovascular disorders in the 6 mo before participation; engagement in occupational tasks involving the upper extremities; and use of pharmacological agents or stimulants (e.g., coca leaf, caffeine-based supplements) commonly used to mitigate symptoms of altitude sickness. All participants provided written informed consent before participation.

Sample Characteristics

Twelve healthy adults (5 males and 7 females) participated in the study. They ascended from sea level to 3,600 m within a 5-h travel period, without intermediate stops or prior altitude exposure. Self-reported sex, height, and weight were recorded, and body mass index (BMI) was calculated. Acute mountain sickness (AMS) was assessed using the Lake Louis Scoring System (LLS; Roach et al. 20), which assigns scores from 0 (no symptoms) to 3 (severe symptoms) for the following AMS symptoms: headache, gastrointestinal symptoms, fatigue/weakness, dizzy/light-headedness, and difficulty sleeping. Oxygen saturation (Sp_{O_2}) was measured using a fingertip pulse oximeter (Biobase Bioindustry Co., Ltd, Jinan, Shandong, PR China) placed on the index finger of the nondominant hand during protocol execution.

Surface Electromyography

Surface EMG sensors were placed following the SENIAM guidelines (21). The skin was shaved and cleaned with an alcohol solution before electrode placement. Participants remained standing while hexagonal Ag/AgCl electrodes (adhesive area 4×2.2 cm, diameter 1 cm, and 2 cm interelectrode distance; Noraxon Corporate) were placed on the dominant upper extremity, defined as the hand used for writing. Electrodes were positioned over the biceps brachii and triceps brachii muscles.

Muscular Oxygen and Total Hemoglobin

Muscular oxygen saturation (SmO_2) and total hemoglobin (THb) were measured using a near-infrared spectroscopy sensor (MOXY, Fortiori Design LLC, Hutchinson, Minnesota). The sensor was placed on the muscular belly of the anterior deltoid, equidistant from the clavicle and its humeral insertion (22). The sensor was secured with medical tape and covered with a compatible light shield to prevent potential ambient light interference. Signals were visually inspected online for validation using the PerfPRO platform (PerfPRO Studio 2021, version 8.04.02).

Protocol

Before each data collection session, the electromyography and near-infrared spectroscopy (NIRS) equipment were calibrated, and signal quality was verified according to manufacturer guidelines. Signal integrity was continuously monitored in real time and subsequently confirmed during offline inspection.

Electromyographic signals were recorded telemetrically using the Noraxon TeleMyo DTS Desk receiver (500 µV amplifier, common mode rejection ratio >100 dB, baseline noise <1 µV root mean square, input impedance >100 MΩ, 16-bit resolution analog-to-digital card, Noraxon Corporate), with a sampling frequency of 1500 Hz. Signals were visualized in real time using the MyoResearch platform (version 3.8, Noraxon Corporate) for online validation. A maximal voluntary contraction (MVC) protocol was conducted for signal normalization.

The repetitive upper extremity task followed the methodology proposed by Slopecki et al. (23). First, the distance from the acromion to the tip of the index finger of the dominant upper extremity was measured. Participants were seated in front of a tablet displaying two targets: one at a proximal distance (30% of upper limb length) and the other at a distal distance (100% of upper limb length). Each target emitted a light signal upon contact by the participant.

Participants performed a repetitive reaching and retrieving task of the proximal and distal targets for 120 s, paced at 1 Hz using a metronome. To avoid dual-task interference, participants completed a familiarization session with the auditory metronome before testing. They were instructed to use the cue exclusively as a pacing reference and not as a performance task. Synchronization with the rhythm was visually monitored before data collection to ensure consistency.

The task was conducted at two time points: within the first 3 h of HH exposure (Acute HH) and after 48 h of continuous exposure (48-h HH). Environmental conditions during data collection remained stable across both time points (i.e., acute exposure and 48-h acclimatization), with ambient temperatures ranging from 11 to 18°C and relative humidity between 10% and 20%. All measurements were performed between 1600 and 2300 during the autumn season.

Data Analysis

Signal processing and CCIs computation were performed using MATLAB R2021b (The MathWorks Inc., Natick, MA).

Signal processing.

EMG signals were filtered offline with a 4th-order Butterworth passband digital filter (cutoff frequencies: 20–450 Hz) (10). Subsequently, a smoothing procedure was applied using a 300-sample root mean square window. Finally, the EMG signals were normalized to the peak MVC amplitude, following the recommendations by Guerrero-Henriquez et al. (24) (Fig. 1).

The normalized signals were segmented into 1-s, nonoverlapping windows based on metronome timing. Each movement cycle was divided into reaching (REA) and retrieving (RET) movement phases and further subdivided into four epochs: 0%–25%; 25%–50%; 50%–75%; and 75%–100%. CCIs were calculated for each epoch.

Cocontraction indices computation.

CCIs were calculated using two established methods:

1) Rudolph and Snyder-Mackler (25):

$$\mathrm{CCI}_1(t) = \frac{\mathrm{Input}_L(t)}{\mathrm{Input}_H(t)}(\mathrm{Input}_L(t) + \mathrm{Input}_H(t)).$$

2) Falconer and Winter (26):

$$\mathrm{CCI}_2(t) = \frac{2 \times \mathrm{Input}_L(t)}{(\mathrm{Input}_L(t) + \mathrm{Input}_H(t))},$$

where, Input_L and Input_H represent the lower and higher EMG values, respectively, between the antagonist muscle pair at time t, normalized to the task cycle. The triceps brachii was defined as the agonist during REA, and the biceps brachii as the agonist during RET. In both cases, higher CCI values indicate increased metabolic costs from the nervous system to stabilize the joint due to motor control deficits (14).

Statistical Analysis

Statistical analyses were performed using SPSS (version 21.0, IBM Corp., Armonk, NY). The significance level was set at $\alpha = 0.05$ for all statistical tests. The Shapiro–Wilk and Levene's tests were used to assess normality and homogeneity of variances, respectively.

Sociodemographic characteristics are reported as medians and interquartile ranges or as means and standard deviations, depending on data distribution. Between-group comparisons by sex (i.e., male, female) and HH exposure time (i.e., acute HH vs. 48-h HH) were conducted using independent t tests or Mann–Whitney U tests, and paired t tests or Wilcoxon tests, as appropriate.

Post hoc power analysis was performed using G*Power 3.1.9.4 (G*Power, Version 3.0.10, Franz Faul, Universität Kiel, Germany) (27), based on Pillai's trace from multivariate tests. The number of movement epochs (four: 0%–25%, 25%–50%, 50%–75%, and 75%–100%), two measures (CCI_1 and CCI_2), and the total number of participants recruited were used as parameters.

A multivariate analysis of variance (MANOVA) model was used to examine the effects of hypobaric hypoxia exposure time (acute vs. 48-h HH), movement phase (reaching vs. retrieving), and movement epoch (0%–25%, 25%–50%, 50%–75%, 75%–100%) on the dependent variables (CCI1 and CCI2). Multivariate effects were assessed using Pillai's trace. Significant multivariate results were followed by univariate ANOVAs to explore effects on each dependent variable. Interaction terms between exposure time, movement phase, and movement epoch were included to evaluate their combined influence on CCIs. Bonferroni adjustments were applied for multiple comparisons. Effect sizes were reported as partial eta squared (η_p^2), with thresholds defined as small ($0.20 \le \eta_p^2 < 0.50$), medium ($0.50 \le \eta_p^2 < 0.80$), and large ($\eta_p^2 \ge 0.80$) (28).

RESULTS

Participant Characteristics

Twelve participants (5 males, 7 females) completed the study. Post hoc power analysis revealed a statistical power of 99% (effect size = 2.58; Pillai's $V = 0.87$). As expected, height and weight differed significantly between sexes; however, BMI did not (Table 1). Acute mountain sickness (AMS) scores were significantly higher at 3 h of HH exposure (acute) compared with 48 h post-arrival (2 [0.75] vs. 0 [0.00]; $P = 0.023$), with no sex differences observed (Table 1).

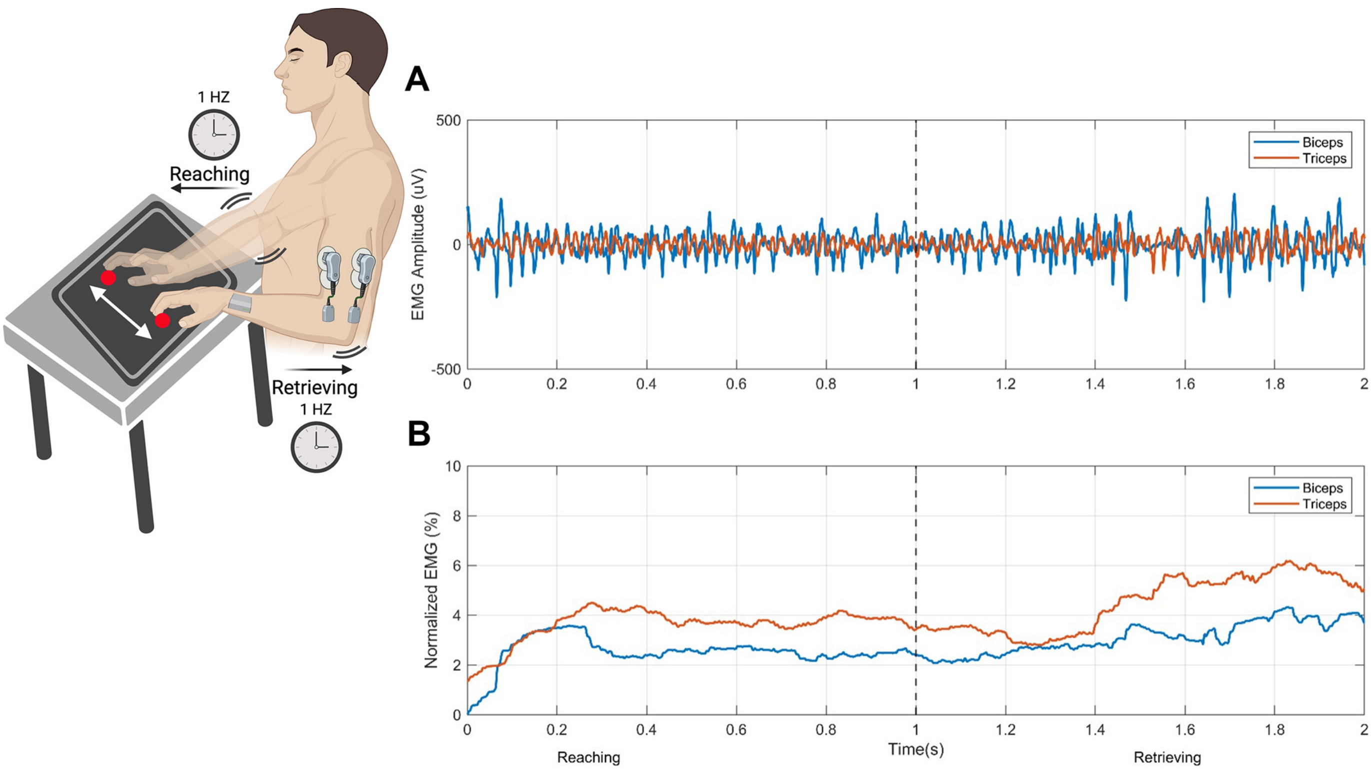

Figure 1. Experimental setup and electromyographic (EMG) signal processing during a repetitive motor task under hypobaric hypoxia. The *left panel* illustrates the experimental setup, including sensor placement on the biceps brachii, triceps brachii, and anterior deltoid, as well as the repetitive reaching-retrieving task paced at 1 Hz. *A*: raw EMG signals: representative unprocessed recordings (in microvolts, μV) of the biceps brachii (*top*) and triceps brachii (*bottom*) during one complete reaching-retrieving cycle. Signal variability reflects natural muscle activation and noise components. *B*: processed and normalized EMG signals: the same signals filtered and normalized to maximal voluntary contraction (% MVC). Smoothed traces represent muscle activation profiles over time. The dashed vertical line indicates the transition from the reaching to the retrieving phase.

Total hemoglobin (THb) was significantly reduced at 48 h of HH exposure compared with the acute condition (12.15 [12.00–12.80] a.u. vs. 12.08 [11.77–12.26] a.u.; $P = 0.010$). Notably, THb values were also significantly lower in females compared with males at both time points (Acute: 12.82 [0.51] a.u. vs. 12.07 [0.23] a.u., $P = 0.010$; 48-h HH: 12.71 [0.80] a.u. vs. 12.01 [0.49] a.u., $P = 0.048$). No significant differences were found in Sp_{O_2} or SmO_2 across sex or exposure conditions (Table 1).

Effects of HH Exposure on Cocontraction Indexes

Exposure to HH significantly increased cocontraction indices (CCI) values ($F_{1,44} = 4.172$; $P = 0.047$; $\eta_p^2 = 0.515$). In contrast, movement phase ($F_{1,44} = 1.777$; $P = 0.189$; $\eta_p^2 = 0.256$)

Table 1. *Participants' characteristics*

Variable	All (n = 12)	Male (n = 5)	Female (n = 7)	*P* Value by Sex	*P* Value by HH
Height, m	1.65 ± 0.12	1.78 ± 0.06	1.56 ± 0.04	0.002*	
Weight, Kg	66.29 ± 11.25	76.00 ± 7.21	59.00 ± 7.30	0.028*	
BMI, Kg/m^2	24.18 ± 1.85	24.08 ± 2.08	24.25 ± 1.99	0.911	
AMS (score)					
Acute HH	2 [0.75]	2 [0.50]	2 [0.20]	1.000	0.023*
48-h HH	0 [0.00]	0 [0.00]	0 [0.00]	1.000	
Sp_{O_2} (%)					
Acute HH	88.90 ± 3.24	89.80 ± 4.14	88.00 ± 2.12	0.413	0.122
48-h HH	87.60 ± 4.35	87.00 ± 5.95	88.20 ± 2.48	0.689	
SmO_2 (a.u.)					
Acute HH	63.07 [11.97]	61.10 [14.45]	65.05 [16.66]	0.755	0.388
48-h HH	66.02 [16.38]	65.34 [16.71]	66.92 [21.26]	0.638	
THb (a.u.)					
Acute HH	12.15 [0.80]	12.82 [0.51]	12.07 [0.23]	0.010*	0.010*
48-h HH	12.08 [0.49]	12.71 [0.80]	12.01 [0.49]	0.048*	

Data are expressed as median [interquartile range] or mean ± standard deviation. AMS, acute mountain sickness; a.u., arbitrary units; BMI, body mass index; HH, hypobaric hypoxia; SmO_2, muscular oxygen saturation; Sp_{O_2}, oxygen saturation; THb, total hemoglobin. *Statistically significant differences ($P < 0.05$).

and movement epoch ($F_{3,44} = 0.023$; $P = 0.068$; $\eta_p^2 = 0.054$) showed no significant effects, and no interactions between these factors were observed.

Univariate analyses confirmed significant differences in both CCI_1 and CCI_2 between acute and prolonged HH exposure. Specifically, CCI_1 values were significantly higher after 48 h of HH exposure across all movement epochs (0%–25%, 25%–50%, 50%–75%, and 75%–100%) in both the reaching and retrieving phases ($P < 0.05$, Fig. 2). Similarly, CCI_2 values also increased significantly in prolonged HH conditions compared with acute exposure in all epochs ($P < 0.05$, Fig. 3).

DISCUSSION

This study investigated the effects of natural hypobaric hypoxia on surface EMG-based cocontraction indices during a repetitive upper extremity task. Two widely used CCI formulations were analyzed (25, 26). The results indicate that 48 h of continuous HH exposure increases CCIs, with medium effect sizes, irrespective of the formulation used. These findings support the hypothesis that HH increases CCI as a compensatory strategy used by the nervous system to stabilize movement.

Although previous studies had demonstrated an increase in EMG amplitude (7–9) under hypoxic conditions, this study adds to the literature by focusing on intermuscular coordination rather than individual muscle behavior. Increased CCI values observed after 48 h may reflect an adaptive response of the nervous system. As noted in studies by Bandini et al. (14) and Sheng et al. (17), increased CCIs values may reflect greater joint stiffness. However, joint stiffness was not directly assessed, and therefore cannot be confirmed as the mechanism.

Due to methodological constraints, this study focused on acute and 48-h HH exposure. Beyond this time window, acclimatization processes may further modulate physiological responses (6). The observed increase in CCIs could reflect an ongoing adaptation of the neuromuscular system to hypoxic stress, characterized by less efficient but more conservative motor strategies (9).

The observed CCI increases may reflect elevated metabolic costs due to motor inefficiency. However, since metabolic stress was not directly assessed across all conditions, we cannot confirm whether this mechanism underlies the observed changes. Mechanistically, electromyographic muscle fatigue, likely driven by elevated intramuscular concentrations of H^+ and lactate in HH conditions (15, 17), may alter neuromotor strategies, promoting greater cocontraction to mitigate the loss of joint stability. Alternatively, this increase in CCIs could result from other strategies used by the nervous system in an inflammatory state (29, 30) and reflect alterations in the synchronization between agonist and antagonist muscles due to a reduction in the efficiency of corticospinal control (17). Alterations in these corticospinal pathways have been correlated to increased agonist-antagonist cocontraction due to enhanced recurrent Renshaw inhibition (31) and decreased Ia reciprocal inhibition (32). The increase in CCIs magnitude after 48 h of

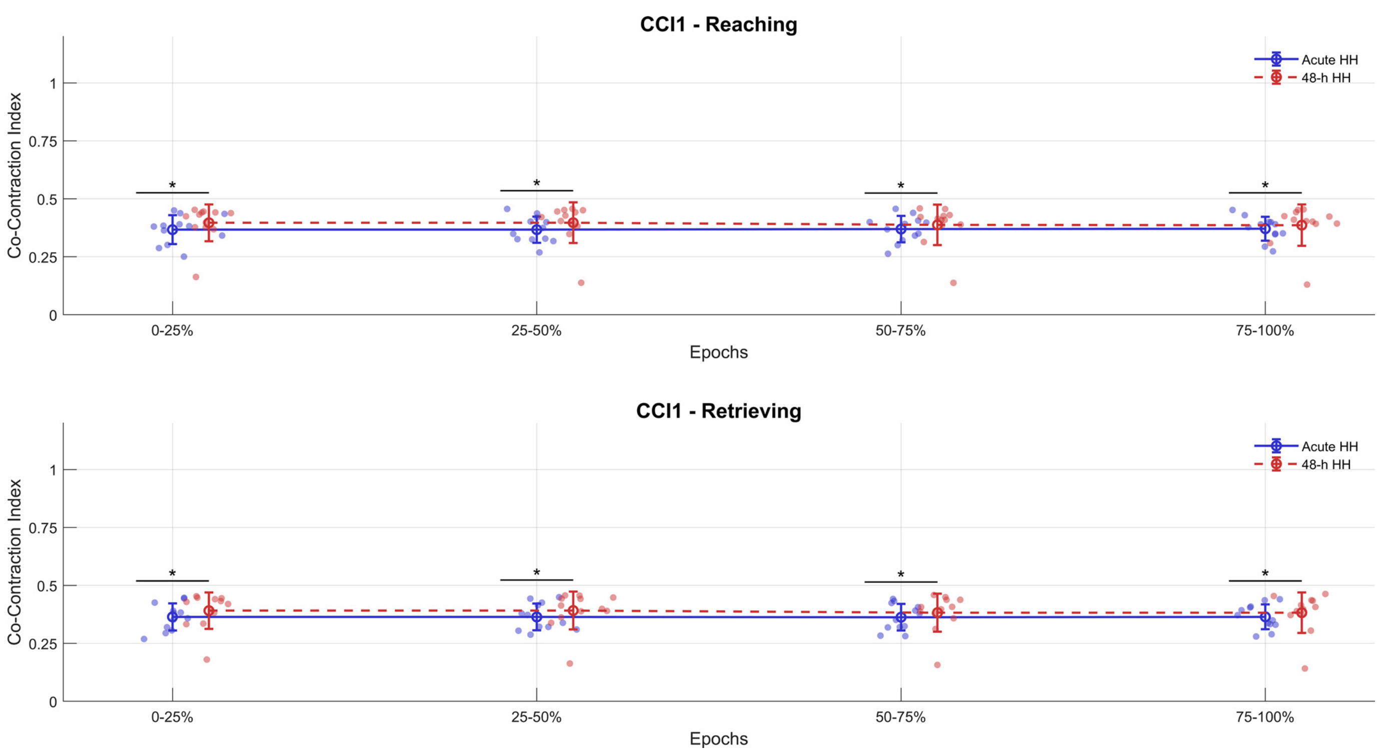

Figure 2. Cocontraction index 1 (CCI1) across movement epochs during reaching and retrieving phases under acute and prolonged hypobaric hypoxia. The *top panel* shows CCI1 values during the reaching phase, and the *bottom panel* shows CCI1 values during the retrieving phase. Each phase is divided into four time epochs (0%–25%, 25%–50%, 50%–75%, 75%–100%). Comparisons are shown between acute hypobaric hypoxia (<3 h; Acute HH) and 48-h continuous exposure (48-h HH). Dots represent individual participant values; error bars indicate means ± standard deviation. *Statistically significant differences between conditions ($P < 0.05$).

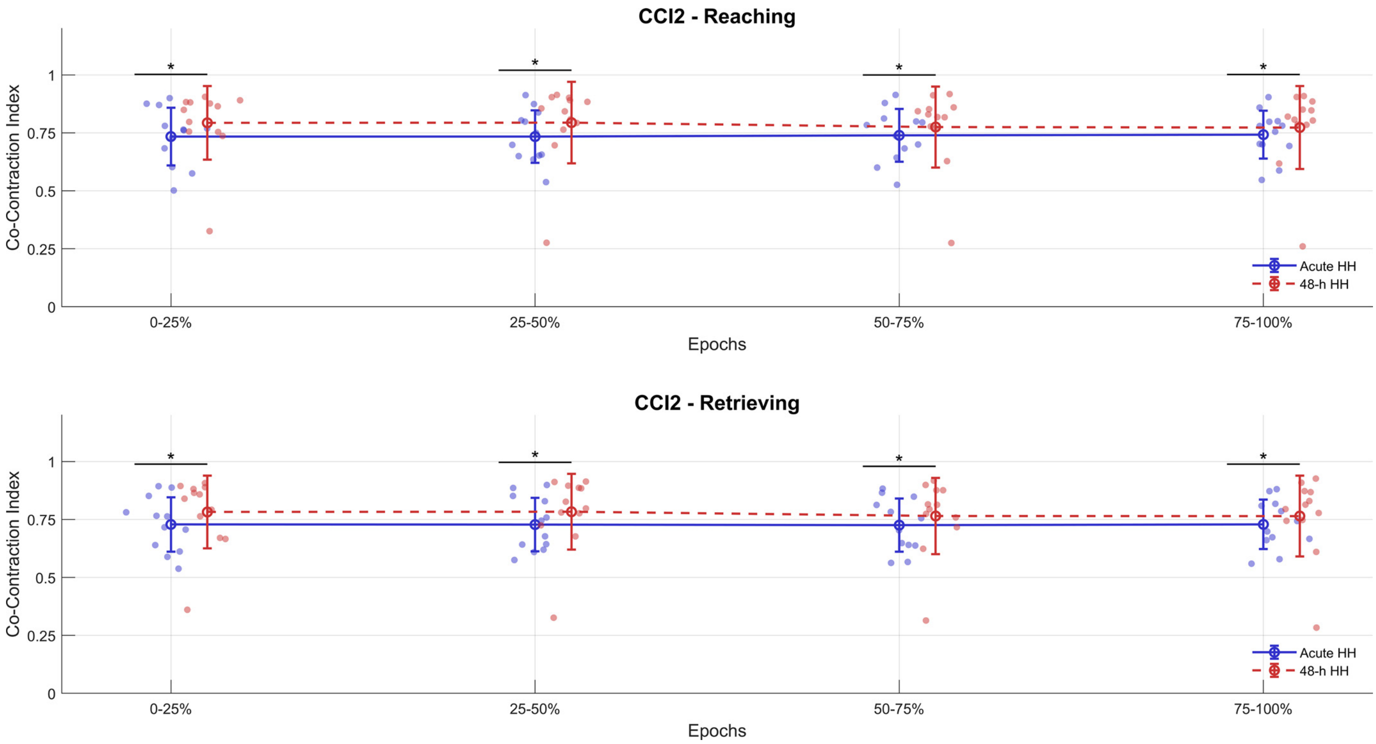

Figure 3. Cocontraction index 2 (CCI2) across movement epochs during reaching and retrieving phases under acute and prolonged hypobaric hypoxia. The *top panel* shows CCI2 values during the reaching phase, and the *bottom panel* shows CCI2 values during the retrieving phase. Each phase is divided into four time epochs (0%–25%, 25%–50%, 50%–75%, 75%–100%). Comparisons are shown between acute hypobaric hypoxia (<3 h; Acute HH) and 48-h continuous exposure (48-h HH). Dots represent individual participant values; error bars indicate means ± standard deviation. *Statistically significant differences between conditions ($P < 0.05$).

HH can be interpreted as an adaptive strategy of the nervous system in the face of HH stress. However, future studies should explore the relationship between these neuromuscular adaptations and long-term motor efficiency, considering diverse exposure protocols and motor tasks.

The proposed mechanisms, such as increased joint stiffness or alterations in corticospinal and inhibitory control, are theoretical and were not directly assessed in this study. These interpretations are supported by prior investigations that reported similar cocontraction patterns in hypoxic and poststroke populations (14, 17). However, given the absence of direct measurements (e.g., stiffness, reflex excitability, or motor-evoked potentials), these mechanisms should be considered plausible hypotheses rather than confirmed explanations of our findings. Future studies should include neurophysiological assessments to clarify the underlying drivers of CCI modulation under hypoxia.

Limitations

The present study has several limitations that may influence the interpretation and generalizability of the results. First, the small sample size and lack of an a priori power calculation designate this study as exploratory. This approach aligns with previous pilot studies that used cocontraction indices in hypoxic conditions with small samples (14, 17). Future research should incorporate larger cohorts and repeated testing sessions to reduce bias and improve external validity.

Second, the within-subject design did not include a normoxic control condition at sea level, which limits the ability to establish causal relationships between hypobaric hypoxia exposure and changes in CCIs. In addition, although higher CCI values are often interpreted as a proxy for increased neuromuscular effort or metabolic cost, alternative mechanisms such as enhanced motor unit recruitment or shifts in oxidative strategy under hypoxic conditions cannot be ruled out. Therefore, caution is warranted when attributing changes in CCI to a single physiological mechanism. To gain deeper insights, future studies should consider longitudinal designs that assess neuromuscular responses over extended time frames and during different stages of acclimatization, ideally incorporating direct measurements of metabolic expenditure.

Third, only two time points were evaluated, acute exposure and after 48 h of continuous exposure, which may not fully capture the dynamic physiological adaptations that occur during a more prolonged acclimatization process. Future studies should consider more extensive follow-up, with intermediate and postextended exposure measurements (>72 h), as well as comparisons between acute and chronic hypoxia.

Sleep quality, though self-reported as good before the 48-h measurement, was not formally assessed. Therefore, potential confounding factors such as undetected sleep disturbances, mild dehydration, or transient peripheral edema could have influenced neuromuscular responses and should be systematically monitored in future studies.

Although all participants were sea-level residents with no prior exposure to altitude, the combination of low AMS scores and relatively preserved oxygen saturation values indicates a favorable individual response to hypoxia or a rapid early-stage adaptation. This could have dampened the magnitude of acute compensatory motor responses, thus limiting the extrapolation of our findings.

Although the menstrual cycle phase of female participants was not recorded, this limitation is unlikely to have influenced the interpretation of our findings. Previous research using near-infrared spectroscopy has shown that total hemoglobin (i.e., THb), the only variable in our study that differed significantly between sexes, remains stable across menstrual phases (33). Nonetheless, future studies should consider controlling for this factor when evaluating sex-based physiological responses.

Finally, the absence of a normoxic control condition at sea level limits the ability to isolate the specific effects of hypoxia on neuromuscular performance. Although our intrasubject design focused on the comparison between acute and 48-h hypoxic exposures, future studies should incorporate baseline assessments at sea level to strengthen causal inference. Moreover, the present protocol was based on a repetitive upper extremity task, which may restrict the extrapolation of our findings to other, more complex motor activities. Future research should consider including a broader variety of motor tasks involving multiple joints and muscle groups to evaluate specific neuromuscular responses across different biomechanical contexts.

Conclusions

A 48-h exposure to hypobaric hypoxia significantly increases surface electromyography-based cocontraction indices (CCIs) values, regardless of the formulation used. These findings suggest higher metabolic costs and a possible reliance on compensatory strategies by the nervous system to maintain joint stability under hypobaric hypoxic stress.

This study provides valuable insights into the neuromuscular mechanisms at play in hypobaric hypoxic environments, highlighting the usefulness of CCIs as effective markers for assessing agonist-antagonist muscle interactions during motor tasks. These results open avenues for further research and practical applications, including altitude training, performance evaluation in extreme conditions, and designing rehabilitation interventions to optimize motor control in hypobaric hypoxic environments.

DATA AVAILABILITY

Data will be made available upon reasonable request.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

J.G.-H. and D.C.A. conceived and designed research; J.G.-H., M.V., and D.C.A. performed experiments; J.G.-H., M.V., M.R.-F., D.A., C.S.-A., and D.C.A. analyzed data; J.G.-H., M.V., M.R.-F., D.A., C.S.-A., and D.C.A. interpreted results of experiments; J.G.-H., M.V., and C.S.-A. prepared figures; J.G.-H. drafted manuscript; J.G.-H., M.V., M.R.-F., D.A., C.S.-A., and D.C.A. edited and revised manuscript; J.G.-H., M.V., M.R.-F., D.A., C.S.-A., and D.C.A. approved final version of manuscript.

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